

Text & speech applications

Sonja Aits

Lund University
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Google Search

I'm Feeling Lucky

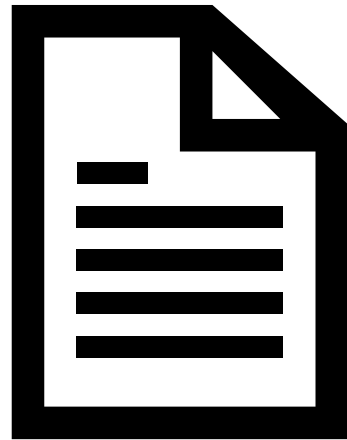
Google offered in: [svenska](#)

Examples of text and speech
applications in medicine and life
sciences?

Regression

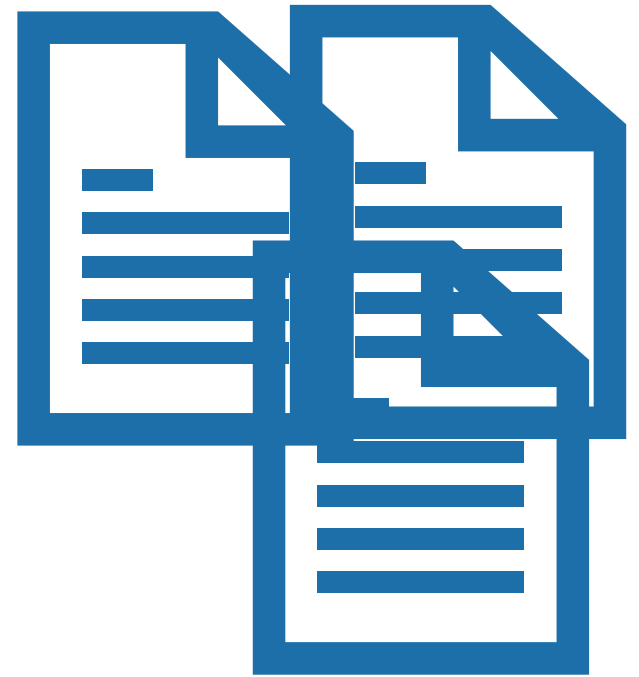
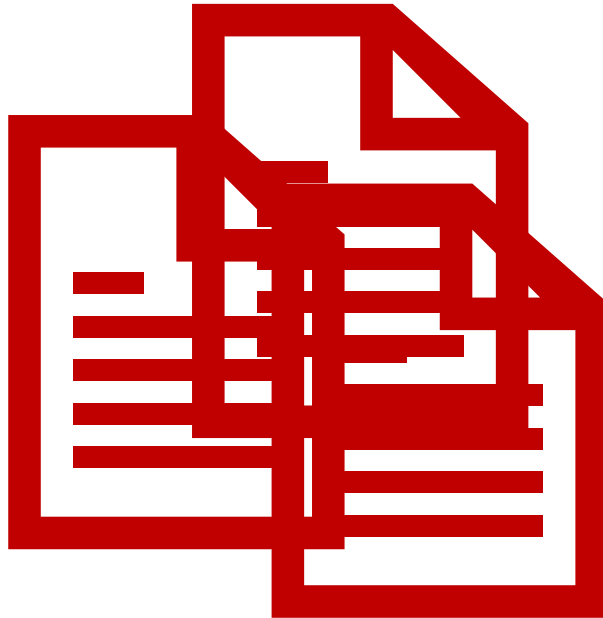
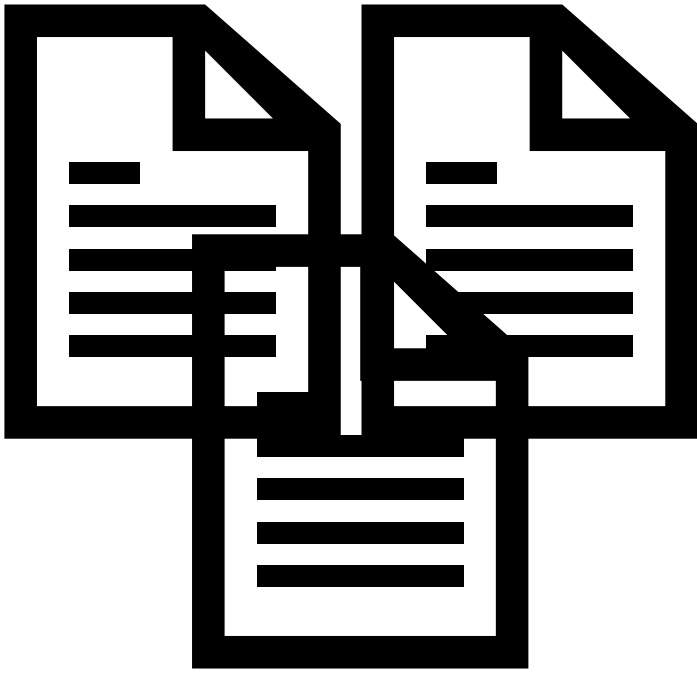
= Predicting a numerical value

Incidence report



Severity score

Clustering



Translation

≡ Google Translate

Text

Documents

DETECT LANGUAGE

ENGLISH

SPANISH

FRENCH



GERMAN

SWEDISH

ENGLISH



Oxidative stress causes lysosomal damage and subsequent cell death in many types of cells.



Oxidativer Stress führt bei vielen Zelltypen zu lysosomalen Schäden und anschließendem Zelltod.



90/5000



Summarization

Risk factors for breast cancer

🌐 3 languages ▾

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From Wikipedia, the free encyclopedia

(Redirected from [Risk factors of breast cancer](#))



This article **needs more [reliable medical references](#) for [verification](#)** or **relies too heavily on [primary sources](#)**. Please review the contents of the article and [add the appropriate references](#) if you can. Unsourced or poorly sourced material may be challenged and [removed](#).

Find sources: ["Risk factors for breast cancer"](#) – [news](#) · [newspapers](#) · [books](#) · [scholar](#) · [JSTOR](#) (November 2016)



Risk factors for breast cancer may be divided into preventable and non-preventable. Their study belongs in the field of [epidemiology](#). Breast cancer, like other forms of cancer, can result from multiple environmental and hereditary risk factors. The term "environmental", as used by cancer researchers, means any risk factor that is not genetically inherited.

For breast cancer, the list of environmental risk factors includes the individual person's development, exposure to [microbes](#), "medical interventions, dietary exposures to nutrients, energy and toxicants, ionizing radiation, and chemicals from industrial and agricultural processes and from consumer products...reproductive choices, energy balance, adult weight gain, body fatness, voluntary and involuntary physical activity, medical care, exposure to tobacco smoke and alcohol, and occupational exposures, including shift work" as well as "metabolic and physiologic processes that modify the body's internal environment."^[1] Some of these environmental factors are part of the physical environment, while others (such as diet and number of pregnancies) are primarily part of the social, cultural, or economic environment.^[1]

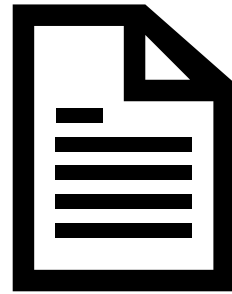
Although many epidemiological risk factors have been identified, the cause of any individual breast cancer is most often unknowable. Epidemiological research informs the patterns of breast cancer incidence across certain populations, but not in a given individual. Approximately 5% of new breast cancers are attributable to hereditary syndromes, and well-established risk factors accounts for approximately 30% of cases.^[2]



Common risk factors for breast cancer are age, female sex, hereditary factors (e.g. BRCA1/2 status) and life style factors (e.g. obesity and alcohol).

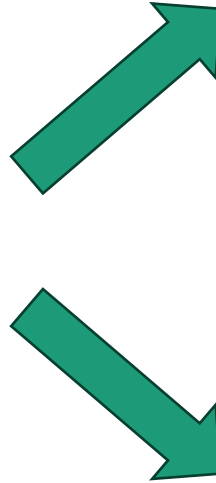
Classification

Electronic
health record



High risk of heart attack

Low risk of heart attack



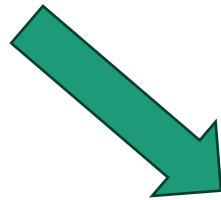
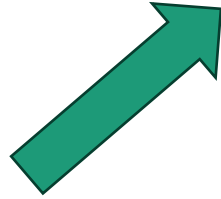
Speech biomarkers

Speech
sample

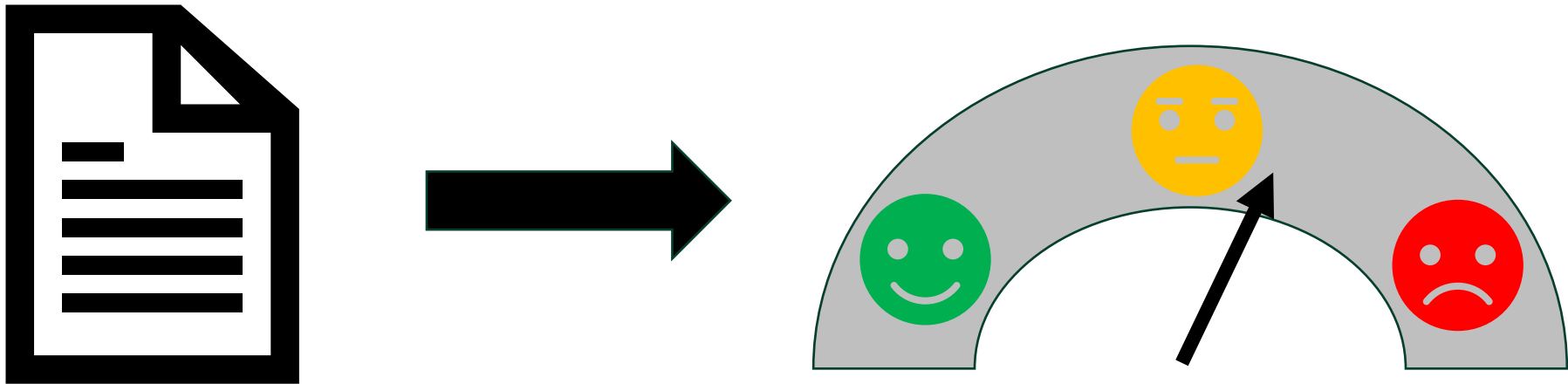


Parkinson's disease

Healthy



Sentiment analysis



Named entity recognition (token classification)

Disease

Drug/treatment

Gene/protein

Process/location

Relation

[PLoS One](#). 2012;7(10):e45381. doi: 10.1371/journal.pone.0045381. Epub 2012 Oct 11.

Identification of cytoskeleton-associated proteins essential for lysosomal stability and survival of human cancer cells.

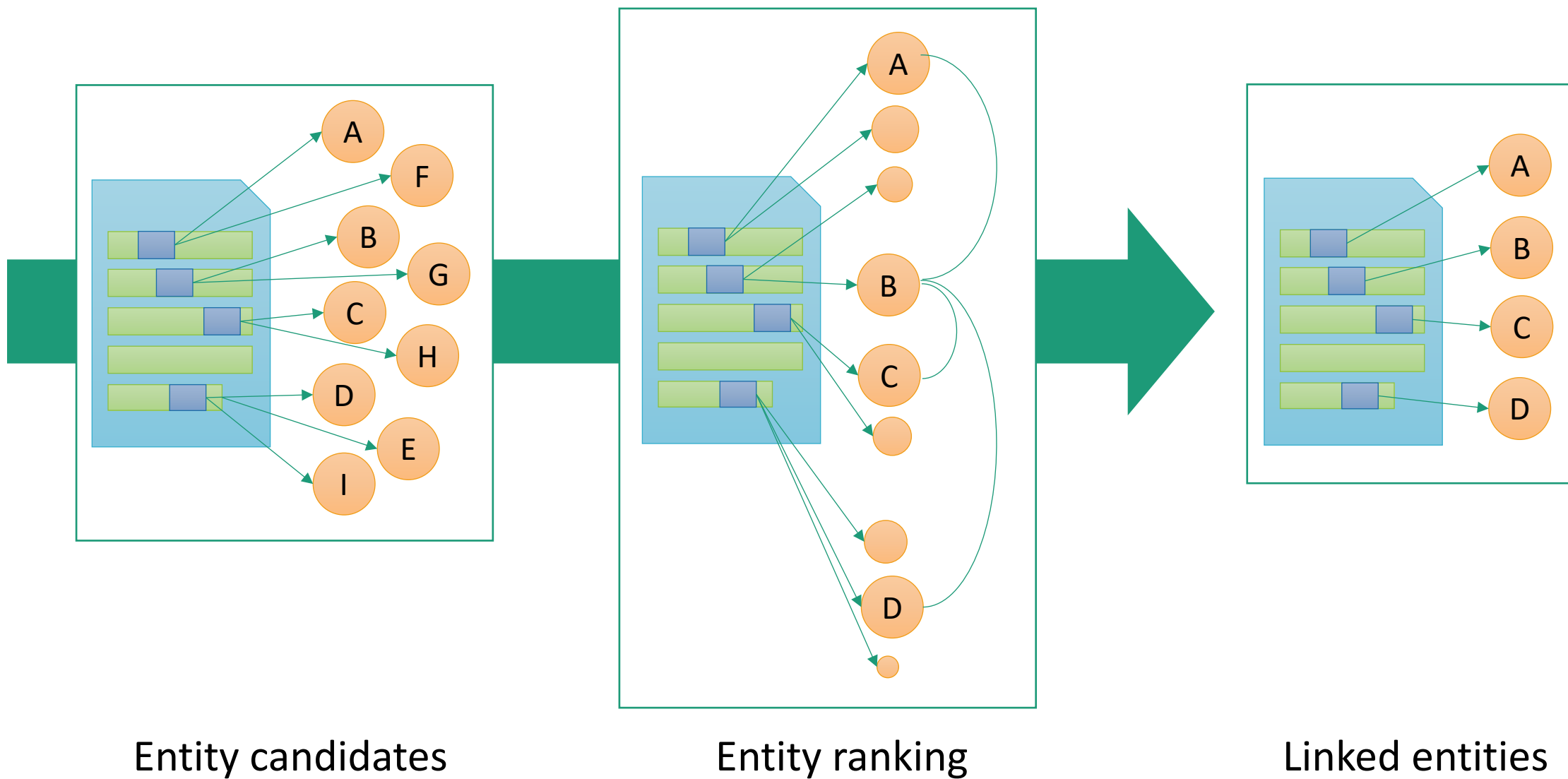
[Groth-Pedersen L](#)¹, [Aits S](#), [Corcelle-Termeau E](#), [Petersen NH](#), [Nylandsted J](#), [Jäättelä M](#).

[Author information](#)

Abstract

Microtubule-disturbing drugs inhibit lysosomal trafficking and induce lysosomal membrane permeabilization followed by cathepsin-dependent cell death. To identify specific trafficking-related proteins that control cell survival and lysosomal stability, we screened a molecular motor siRNA library in human MCF7 breast cancer cells. SiRNAs targeting four kinesins (KIF11/Eg5, KIF20A, KIF21A, KIF25), myosin 1G (MYO1G), myosin heavy chain 1 (MYH1) and tropomyosin 2 (TPM2) were identified as effective inducers of non-apoptotic cell death. The cell death induced by KIF11, KIF21A, KIF25, MYH1 or TPM2 siRNAs was preceded by lysosomal membrane permeabilization, and all identified siRNAs induced several changes in the endo-lysosomal compartment, i.e. increased lysosomal volume (KIF11, KIF20A, KIF25, MYO1G, MYH1), increased cysteine cathepsin activity (KIF20A, KIF25), altered lysosomal localization (KIF25, MYH1, TPM2), increased dextran accumulation (KIF20A), or reduced autophagic flux (MYO1G, MYH1). Importantly, all seven siRNAs also killed human cervix cancer (HeLa) and osteosarcoma (U-2-OS) cells and sensitized cancer cells to other lysosome-destabilizing treatments, i.e. photo-oxidation, siramesine, etoposide or cisplatin.

Named entity linking



Relation extraction

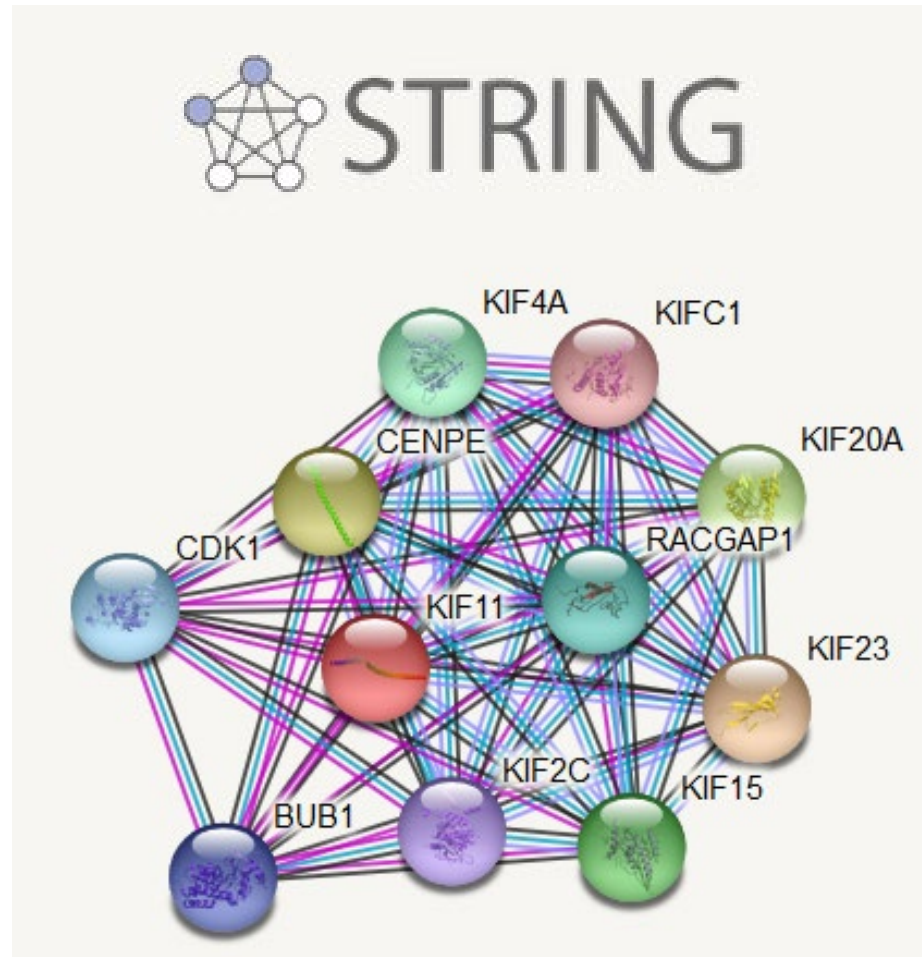
**Rapamycin inhibits mTOR, and
thereby affects autophagy.**

Entity 1: Rapamycin

Entity 2: mTOR

Relation: inhibition

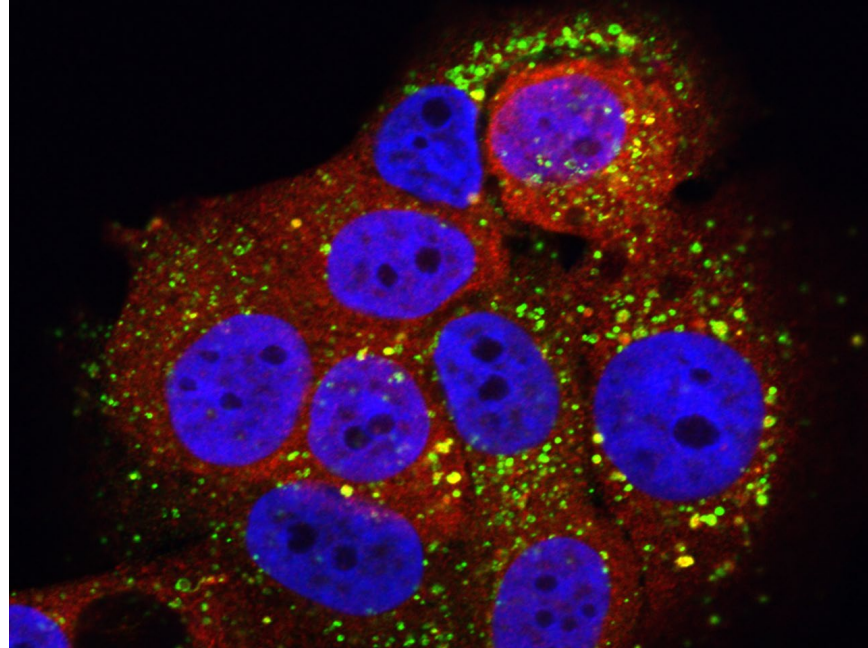
Relation extraction is used to populate protein-protein interaction databases



Data generation

= Producing new data based on existing data

Image captioning

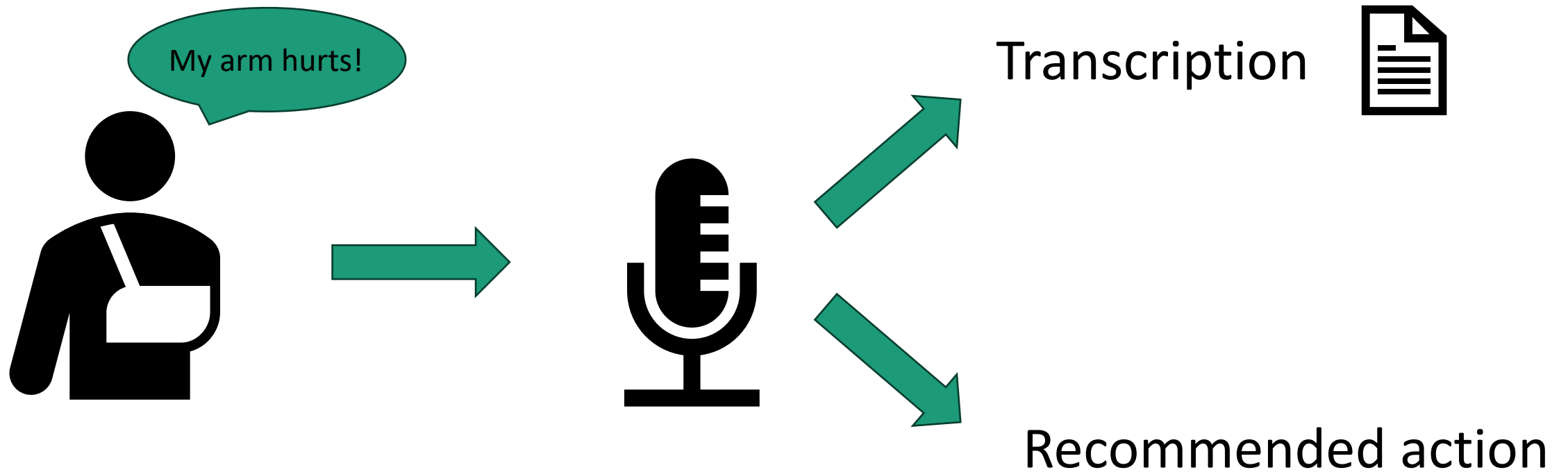


“a fluorescence microscopy image showing a group of cells with blue nuclei and bright green cytosolic spots”

Reasoning

= Making decisions or drawing conclusions based on available information and logic

Speech recognition



General-purpose chatbots



You

what are the most common nlp tasks for deep neural networks?



ChatGPT

Natural Language Processing (NLP) involves a variety of tasks that can be tackled using deep neural networks. Some of the most common NLP tasks for deep neural networks include:

1. **Text Classification:**

- **Task:** Assigning predefined categories or labels to a given text.
- **Example:** Spam detection, sentiment analysis, topic categorization.

2. **Named Entity Recognition (NER):**

- **Task:** Identifying and classifying entities (such as names of people, organizations, locations) in text.
- **Example:** Extracting names of people and locations from news articles.

AI chatbots can execute a broad variety of tasks

- Writing assistant
 - Title suggestions
 - Content suggestions
 - Outline generation
 - Rephrasing/Correction/Shortening
 - Summarization/Style change/Translation
 - Production of complete text
- Discussion partner
- Personal teacher
- Programming assistant
- Personal assistant
- Data analyst
- Image designer
- Software developer
- ...

Virtual medical assistants

I cannot smile or
raise my right arm.



You probably have
a stroke. I am calling 112.



Summary: common tasks for NLP/speech processing

- There is a large variety of AI tasks related to text and speech data
- Multi-modal AI combines multiple data modalities (e.g. images and text)
- AI chatbots are multi-modal general-purpose AI tools